

Airline Performance & Delay Analysis

An interactive dashboard project analyzing flight delays, airline efficiency, and route performance using real-world aviation data.

TABLEAU / POWER BI



Made with GAMMA

Introduction to the Project

This project develops an **interactive dashboard** to analyze airline on-time performance and flight delay patterns, transforming raw aviation data into meaningful visual insights.

Data Source

Kaggle Airline On-Time Performance dataset with real departure and arrival records

Approach

Data cleaning, KPI design, and multi-page dashboard development

Goal

Support data-driven decisions and uncover operational inefficiencies



Problem Statement



Background

Airlines struggle to maintain on-time performance due to **operational inefficiencies, weather disruptions, and high air traffic volume**. Limited visibility into delay patterns makes root cause analysis difficult.

Objective

Design an interactive dashboard to analyze flight delays, measure airline performance, and evaluate route efficiency — enabling faster, evidence-based operational decisions.

Objectives of the Project

1

Measure On-Time Performance

Track average arrival/departure delays across airlines and time periods

2

Identify Delay Patterns

Uncover seasonal trends, peak delay windows, and route-level bottlenecks

3

Analyse Route Complexity

Evaluate distance vs. delay relationships and most problematic routes

4

Enable Interactivity

Allow users to filter by airline, date, and airport for tailored insights

Tools & Technologies Used



Excel / CSV

Primary data source for raw flight records. Used for initial data exploration and formatting before import.



Power BI / Tableau

Building multi-page interactive dashboards with KPI cards, maps, line charts, and drill-through filters.

Data Cleaning & Preprocessing

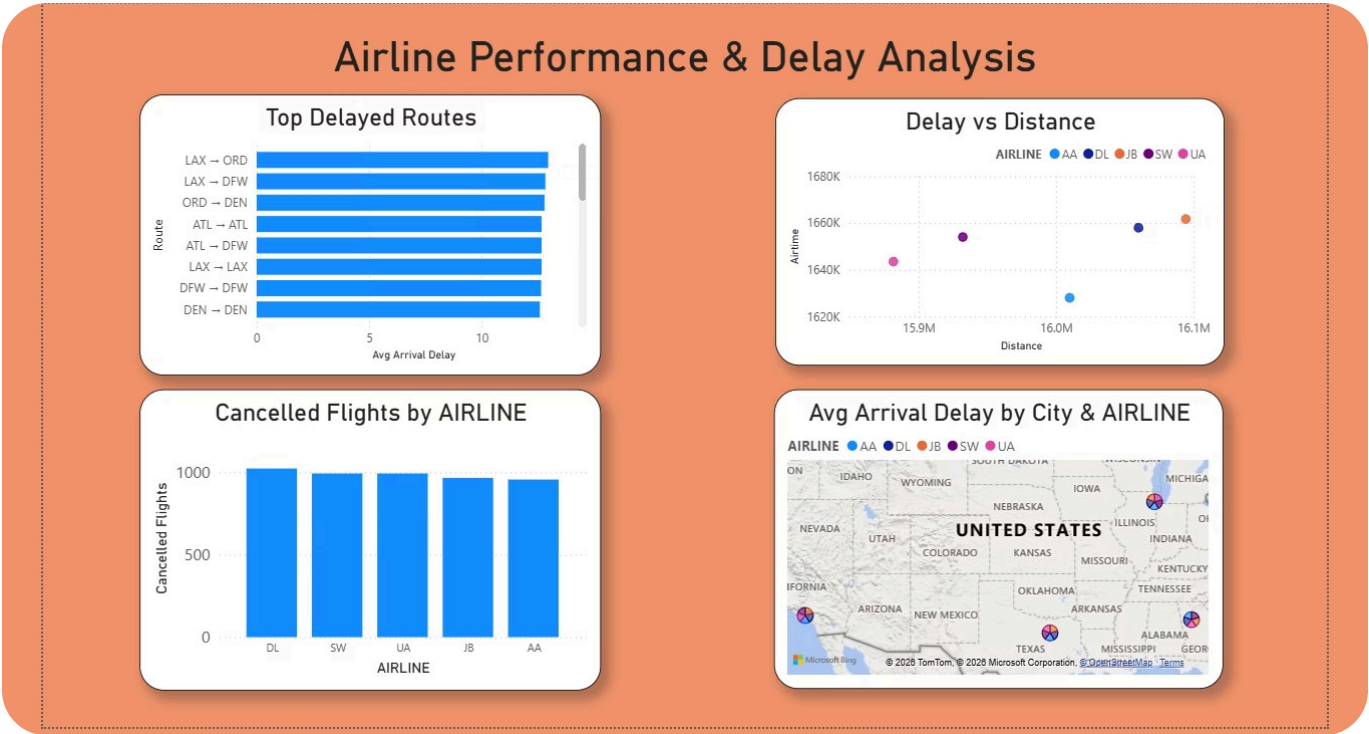


A structured preprocessing pipeline ensured data quality and consistency before visualisation.

Key Cleaning Steps

- **Missing values** handled by filtering or imputing relevant columns
- **Irrelevant columns** removed to optimise dashboard performance
- **Data types** standardised – dates, times, and numeric delay fields

Page 1: Flight Operations Overview



Key Metrics at a Glance

Monthly arrival delay trends and per-airline average delays provide a macro view of overall operational health.

25K

Total Flights

1,747

On-Time Flights

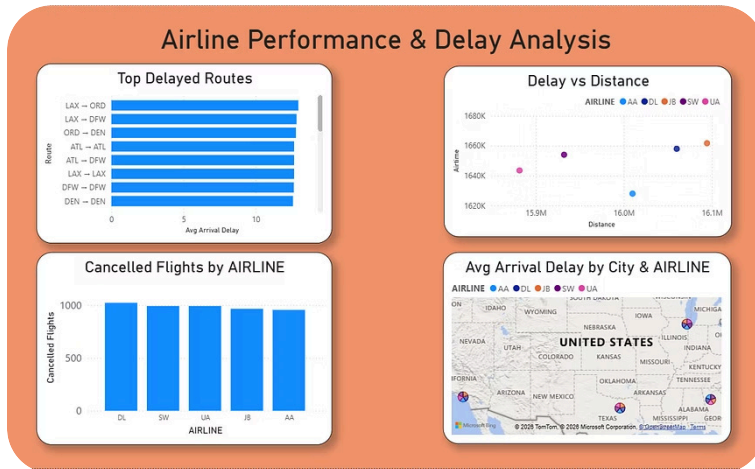
23K

Delayed Flights

149 min

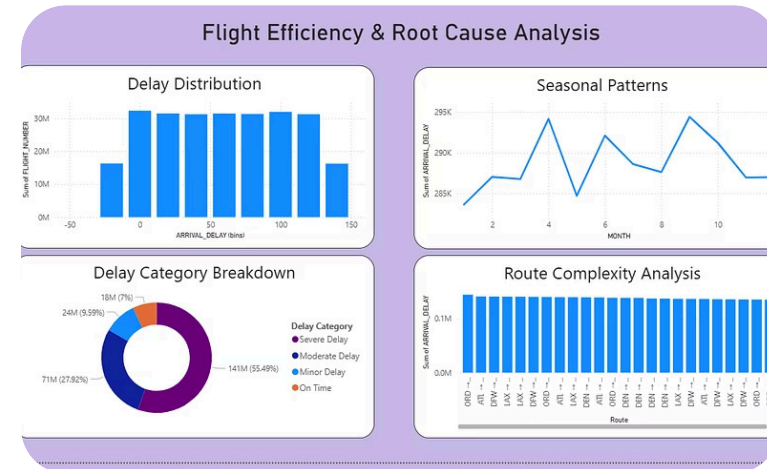
Max Delay

Page 2 & 3: Performance & Root Cause



Airline Performance & Delay Analysis

Top delayed routes, delay vs. distance scatter, cancelled flights by airline, and a geographic map of avg arrival delay by city.



Flight Efficiency & Root Cause Analysis

Delay distribution, seasonal patterns, delay category breakdown (donut chart), and route complexity analysis across airlines.



Key Insights & Findings

High Delay Rate

Over **92% of flights experienced some delay**, with only 1,747 of 25K flights arriving on time — signalling a systemic performance issue.

Seasonal Peaks

Delays spike during **summer and holiday periods**, revealing strong seasonal demand pressure on capacity and scheduling.

Route Complexity Impact

Longer, more complex routes show **higher average delays**, suggesting scheduling buffers are insufficient for high-distance corridors.

Airline Variance

Significant performance differences exist between carriers — some airlines consistently outperform others on the same routes.

Business Recommendations & Conclusion

Recommendations

-  **Optimise scheduling buffers** on high-delay routes, especially during peak seasons
-  **Benchmark low-performing airlines** against top performers to identify operational gaps
-  **Implement real-time monitoring** dashboards for proactive delay intervention

Conclusion

This project demonstrates how **data visualisation transforms raw aviation data** into actionable intelligence. The interactive dashboards reveal delay patterns, highlight underperforming routes, and empower airline operations teams to make faster, evidence-based decisions.

The project aligns with real-world analytics applications and showcases strong skills in **data cleaning, KPI design, and visual storytelling**.